

Side by Side

	Matrix (Thomas)	Feynman-Kac MC	Q-Learning
Needs model?	Yes — exact equations	Partial — just rates	No — blind trial/error
How Z computed	Solve $AZ = b$ directly	Average $e^{\beta T}$ over SSA runs	Not computed — replaced by Q^*
Accuracy	Exact	Noisy (improves with runs)	Approximate (discrete actions)
Speed	Instant	Slow (many runs per state)	Slow (many episodes)
Scales to many species?	No (N^d matrix)	Yes	Yes
Martingale involved?	Proof only	Proof + algorithm	Not directly
k^+ formula	$k^0 \cdot Z(n \pm 1)/Z(n)$	$k^0 \cdot Z(n \pm 1)/Z(n)$	$k^0 \cdot m[\arg \max Q]$

Step 1: You Want to Model Uncertainty

Your policy at state $n_A = 8$ says: "I think the best dose ratio is around $\mu = 0.7$."

But you're not 100% sure. Maybe 0.65 is better. Maybe 0.78. You want to express this uncertainty as a probability distribution over possible actions.

You need a distribution that:

- Is centered at your best guess μ
- Spreads out symmetrically around it
- Has a width controlled by σ (how uncertain you are)

The natural choice for this is the **Normal / Gaussian distribution**.

Step 2: Build It From Scratch

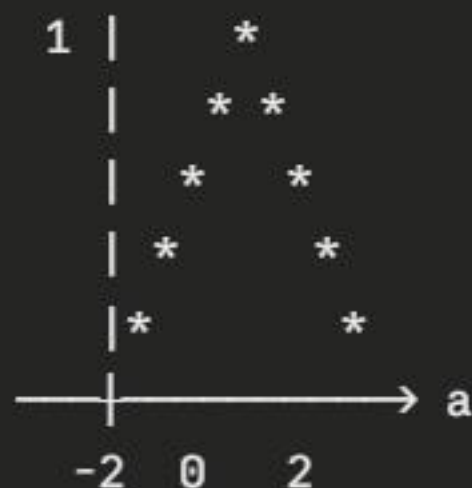
Start with the simplest bump function centered at zero:

$$f(a) = e^{-a^2}$$

Why e^{-a^2} ? Because:

- At $a = 0$: $f(0) = e^0 = 1$ — peak at center
- As a increases: a^2 increases $\rightarrow e^{-a^2}$ decreases $\rightarrow$ tails go to zero
- Symmetric: $f(-a) = f(a)$

Looks like:



Step 3: Add the Mean μ

Shift the center from 0 to μ by replacing a with $(a - \mu)$:

$$f(a) = e^{-(a-\mu)^2}$$

Now peak is at $a = \mu$ instead of $a = 0$.

Step 4: Add Width Control σ

Currently the width is fixed. You want to control how spread out it is.

Replace $(a - \mu)^2$ with $\frac{(a-\mu)^2}{2\sigma^2}$:

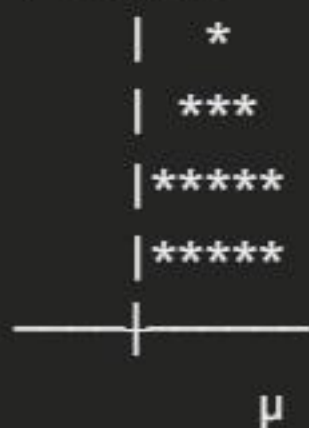
$$f(a) = e^{-\frac{(a-\mu)^2}{2\sigma^2}}$$

Why $2\sigma^2$ and not just σ ?

- σ small $\rightarrow$ denominator small $\rightarrow$ exponent large negative $\rightarrow$ drops off fast $\rightarrow$ **narrow peak**
- σ large $\rightarrow$ denominator large $\rightarrow$ exponent small negative $\rightarrow$ drops off slow $\rightarrow$ **wide spread**

The factor 2 is a convention that makes the math clean later (derivatives work out nicely).

σ small:



σ large:



Step 5: Make It a Proper Probability Distribution

A probability distribution must integrate to 1:

$$\int_{-\infty}^{\infty} P(a) da = 1$$

Currently $\int e^{-\frac{(a-\mu)^2}{2\sigma^2}} da \neq 1$.

What does it equal? This is a famous integral. The answer is:

$$\int_{-\infty}^{\infty} e^{-\frac{(a-\mu)^2}{2\sigma^2}} da = \sigma\sqrt{2\pi}$$

So divide by this to normalize:

$$P(a) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{(a-\mu)^2}{2\sigma^2}}$$

Now it integrates to 1. This IS the Gaussian formula. Nothing more mysterious than: take a bump function, control its center and width, normalize it.

$$\pi_{\theta}(a | s) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(a - \mu_{\theta}(s))^2}{2\sigma^2}\right)$$

Where $\mu_{\theta}(s) = \theta_1 \cdot n_A + \theta_2$ is your linear policy.

Step 6: Why Log of This?

In the policy gradient you need $\log \pi_{\theta}(a \mid s)$.

$$\log \pi_{\theta}(a \mid s) = \log \left(\frac{1}{\sigma \sqrt{2\pi}} \right) + \log \left(e^{-\frac{(a-\mu)^2}{2\sigma^2}} \right)$$

$$= -\log(\sigma \sqrt{2\pi}) - \frac{(a - \mu)^2}{2\sigma^2}$$

The log turned the product into a sum, and $\log e^x = x$ killed the exponential. Much simpler to work with.

The first term $-\log(\sigma \sqrt{2\pi})$ is a constant — doesn't depend on θ . Gradient = 0.

Only the second term matters:

$$\log \pi_{\theta}(a \mid s) \propto -\frac{(a - \mu_{\theta}(s))^2}{2\sigma^2}$$

Now — Full Algorithm All At Once

Here's every step of REINFORCE for the Moran problem in one place:

Setup

State: $s = n_A$ Actions: $a \in \mathbb{R}$ (dose ratio)

Policy: $\mu_\theta(n_A) = \theta_1 n_A + \theta_2$ $\sigma = 0.3$ (fixed)

$$k_1^\dagger = k_1^0 \cdot a \quad k_2^\dagger = k_2^0 / a$$

Reward: $r_t = +1$ per step alive, 0 at absorption

Step A — Initialize

$$\theta_1 = 0, \quad \theta_2 = 1, \quad \alpha = 0.001, \quad \gamma = 0.97$$

(means: start with uncontrolled policy — flat line at 1)

Step B — Collect One Episode

From n_A^0 , run until absorption:

At each step t :

$$\mu_t = \theta_1 \cdot n_{A,t} + \theta_2$$

$$a_t \sim \mathcal{N}(\mu_t, \sigma^2) \quad \leftarrow \text{sample action}$$

$$k_1^\dagger = k_1^0 \cdot a_t, \quad k_2^\dagger = k_2^0 / a_t$$

Run SSA step $\rightarrow$ new state $n_{A,t+1}$, reward r_t

Collect: $(s_0, a_0, r_0), (s_1, a_1, r_1), \dots, (s_T, a_T, r_T)$

Step C — Compute Returns

Work backwards from end of episode:

$$R_T = r_T$$

$$R_{T-1} = r_{T-1} + \gamma R_T$$

$$R_{T-2} = r_{T-2} + \gamma R_{T-1}$$

$$\vdots$$

$$R_t = r_t + \gamma R_{t+1}$$

Each R_t = total discounted reward from step t onward.

Step D — Compute Gradient

$$g_{\theta_1} = \sum_{t=0}^T \underbrace{\frac{(a_t - \mu_t) \cdot n_{A,t}}{\sigma^2}}_{\nabla_{\theta_1} \log \pi} \cdot \underbrace{R_t}_{\text{return}}$$

$$g_{\theta_2} = \sum_{t=0}^T \underbrace{\frac{(a_t - \mu_t)}{\sigma^2}}_{\nabla_{\theta_2} \log \pi} \cdot \underbrace{R_t}_{\text{return}}$$

Reading this: if action $a_t > \mu_t$ (sampled above mean) AND R_t is large (good episode) $\rightarrow$ gradient pushes θ to increase μ at this state $\rightarrow$ mean shifts up $\rightarrow$ more likely to take higher doses here next time.

Step E — Update Parameters

$$\theta_1 \leftarrow \theta_1 + \alpha \cdot g_{\theta_1}$$

$$\theta_2 \leftarrow \theta_2 + \alpha \cdot g_{\theta_2}$$

Step F — Repeat B through E

Many episodes. θ_1 drifts negative, θ_2 adjusts. After convergence:

$$\mu_{\theta}(n_A) = \theta_1 n_A + \theta_2$$

Should match the PRX curve $k_1^{\dagger}/k_1^0$ vs n_A :

- Low n_A : $\mu > 1 \rightarrow$ boost A
- $n_A = n^*$: $\mu = 1 \rightarrow$ OFF mode
- High n_A : $\mu < 1 \rightarrow$ suppress A